

Duplicate and Arrange Script (MJ)

Script Name:

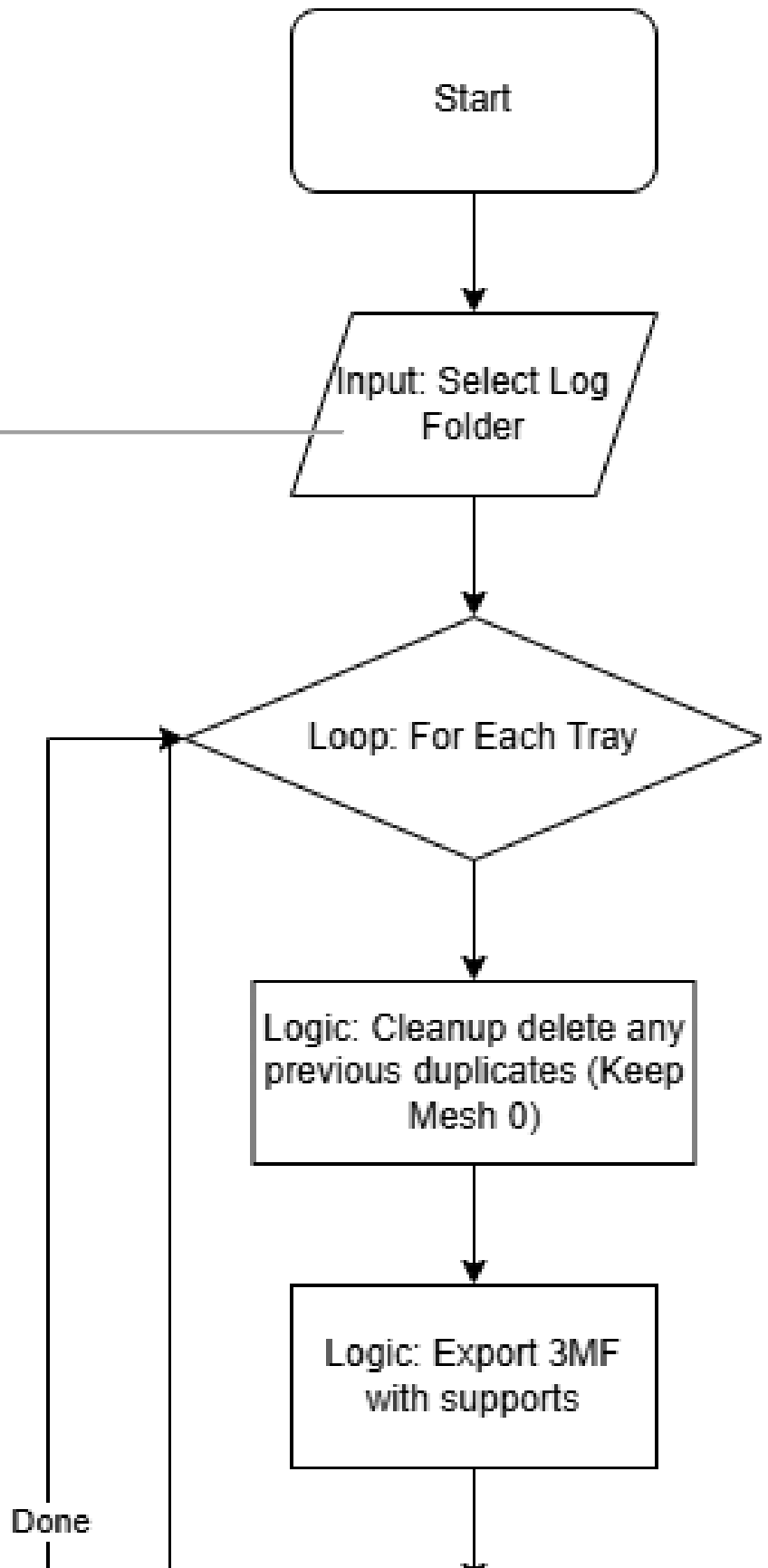
duplicate_and_arrange_MJ.lua

Functionality

This script automates the production workflow by filling trays with duplicate parts until a specified density (60%) is reached. It ensures that support structures are preserved during duplication and efficiently packs the tray using a scanline algorithm.

Logic Flow:

- **Cleanup:** Removes all existing duplicates in the tray, keeping only the first mesh (Index 0) as the template.
- **Calculation:** Estimates the part's footprint (Area) and the tray's available area (assuming a cylindrical platform if configured). It calculates how many copies are needed to reach the target fill percentage.



- **Duplication:** Since standard duplication loses support data, the script exports the template part (with supports) to a temporary 3MF file, then iteratively imports it back into the tray to create copies with supports intact.
- **Packing:** Uses the 2D Scanline Packer to arrange all parts within the build platform.

User Guide (Quick Start)

1. Ensure your tray contains exactly **one** instance of the part you want to mass-produce, with supports already applied.
2. Run the script **duplicate_and_arrange_MJ**.
3. Select a folder for the log files when prompted.
4. The script will automatically duplicate the part to fill 60% of the tray area and pack them.
5. Wait for the "Arrangement Complete" popup.

Code Maintenance & Technical Notes

Support Preservation: The `luamesh:dupe()` function does not copy support dependencies. The script solves this by splitting supports into a separate mesh, exporting both to a 3MF, and re-importing them. **Do not modify this export-import loop** without understanding the `assignsupport` mechanism.

Packing Algorithm: The script uses **Scanline Packing** (``packingid_2d``). Trueshape packing is avoided because it ignores "No-Build" zones, which causes collisions in cylindrical build volumes (e.g., Hyrax).

Cylindrical Math: The variable ``is_cylinder`` (set to ``true``) forces the tray area calculation to use ``pi * r^2``. Netfabb's internal bounding box logic sees the machine as a square, so this manual calculation is critical for accurate fill estimation.